Digital Twin of Astronaut Heart Rate & Oxygen Response During Light

Overview

This project presents a Digital Twin system designed to simulate and analyze the physiological response of an astronaut during different stages of space flight. The main focus of the project is monitoring Heart Rate (HR) and Oxygen Saturation (SpO2) changes during launch, ascent, microgravity transition, and stabilization phases.

The system uses the BioGears Physiology Engine to generate realistic biomedical simulation data and applies digital twin concepts to observe astronaut health conditions in real time.

The project helps understand how human physiology reacts under varying gravitational and environmental conditions during space missions.

Objectives

- Simulate astronaut physiological behavior during flight.
- Monitor heart rate variations during launch and microgravity.
- Analyze oxygen response under changing environmental conditions.
- Build a Digital Twin model for astronaut health tracking.
- Visualize physiological data using graphs and dashboards.

Features

- Real-time physiological simulation using BioGears.
- Heart Rate and Oxygen Saturation monitoring.
- Mission phase-based analysis.
- Automatic graph generation.
- CSV-based simulation output.
- Interactive visualization dashboard.
- Statistical analysis of physiological parameters.

Technologies Used

Technology	Purpose
Python	Core programming language
BioGears Engine	Physiological simulation
NumPy	Numerical computations
Pandas	Data processing

Matplotlib	Data visualization
HTML / Chart.js	Dashboard visualization

System Architecture

```
BioGears Simulation Engine
            ↓
Physiological Data Generation
            ↓
Digital Twin Processing Layer
            ↓
Heart Rate & SpO2 Analysis
            ↓
Visualization Dashboard
            ↓
Reports & Statistics
```

Project Structure

```
project/
├── biogears/
│   ├── bin/
│   ├── patients/
│   ├── scenarios/
│   └── environments/
│
├── results/
│   ├── astronaut_data.csv
│   ├── plots/
│   ├── statistics.json
│   └── logs/
│
├── run_simulation.py
├── analyze_results.py
├── dashboard.html
├── requirements.txt
└── README.md
```

Mission Phases

1. Pre-Launch Phase

- Normal Earth gravity.
- Stable heart rate and oxygen levels.

2. Launch Phase

- Increased G-force stress.
- Rapid increase in heart rate.
- Temporary oxygen fluctuation.

3. Ascent Phase

- Transition from atmosphere to space.
- Cardiovascular adaptation begins.

4. Microgravity Phase

- Reduced gravitational pressure.
- Physiological stabilization.
- Oxygen response adjustment.

5. ISS Stabilization Phase

- Stable physiological condition.
- Heart rate gradually normalizes.

Physiological Parameters Monitored

- Heart Rate (HR)
- Oxygen Saturation (SpO2)
- Blood Pressure
- Respiration Rate
- Cardiac Output
- Core Body Temperature
- Stress Hormone Levels

How the Digital Twin Works

The Digital Twin continuously mirrors astronaut physiological conditions using simulation data from BioGears. The system compares real-time values with normal physiological ranges and identifies stress patterns during different flight stages.
The model helps:

- Predict abnormal physiological conditions.

- Monitor astronaut health status.
- Analyze cardiovascular adaptation.
- Support future space healthcare systems.

Installation

Step 1: Clone the Repository

```
git clone <repository_link>
cd project
```

Step 2: Install Dependencies

```
pip install -r requirements.txt
```

Running the Project

Run the Simulation

```
python run_simulation.py
```

Run Data Analysis

```
python analyze_results.py
```

Open Dashboard

```
python -m http.server 8080
```

Then open:

```
http://localhost:8080
```

Sample Results

Heart Rate Observation

- Significant increase during launch.
- Gradual stabilization in microgravity.

Oxygen Saturation Observation

- Temporary reduction during ascent.
- Stable oxygen adaptation inside ISS environment.

Applications

- Space healthcare monitoring.
- Astronaut physiological research.
- AI-based medical prediction systems.
- Future Mars mission health analysis.
- Biomedical Digital Twin research.

Future Improvements

- Real-time sensor integration.
- AI-based anomaly detection.
- Advanced predictive analytics.
- Machine learning health scoring.
- Extended deep-space simulation.

Conclusion

This project demonstrates how Digital Twin technology can be applied to astronaut physiological monitoring during space missions. By analyzing heart rate and oxygen response, the system provides insights into human adaptation under extreme flight conditions.

The project combines biomedical simulation, data analysis, and visualization techniques to build a foundation for future intelligent astronaut healthcare systems.

Contributors

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License

This project is developed for academic and research purposes.